FBISE Physics Class 11

Complete Step-by-Step Solutions | 48 Past-Paper Questions

Made by Fakhar STEM Sphere

Questions first. Full working after every question.

This study booklet follows the order of the supplied **FBISE Physics 1st Year Most Important Past Paper Numericals**. It includes all 48 items, including the conceptual questions and both versions of the strain-energy problem.

Each answer preserves the detailed method explained in the chat: **Given Data, To Find, Formula Used, rearrangement, substitution, calculation, final units, and the explanation needed to understand the method.**

The supplied **Formula Sheet** has been used as a reference. Where a required relationship is absent from the sheet, it is written in the solution. Missing data and printing errors are identified rather than silently replaced.

How to write a numerical in the exam

1. Write the known quantities with symbols and units.
2. Convert the data to suitable units, usually SI units.
3. State what is required.
4. Write the correct physical formula and rearrange it.
5. Substitute values and show the calculation.
6. Box the final answer with its unit and direction where needed.

Use extra digits during the calculation and round the final answer. The level of detail here is intended for learning; in the exam, retain the essential working clearly.

Constants used

$$g = 9.8\ \mathrm{m\,s^{-2}}, \qquad e = 1.6 \times 10^{-19}\ \mathrm{C}$$

$$k = 9.0 \times 10^{9}\ \mathrm{N\,m^2\,C^{-2}}, \qquad c = 3.0 \times 10^{8}\ \mathrm{m\,s^{-1}}$$

$$m_p = 1.673 \times 10^{-27}\ \mathrm{kg}$$

Temperature convention: Use $T(\mathrm{K}) = t(^\circ\mathrm{C}) + 273$ for these school-level calculations.

Notation: u means initial speed or initial velocity component; v means final speed or the velocity indicated by the context. KE is kinetic energy and PE is gravitational potential energy.

1. Water rocket: speed, height, and energies

Question. In an experiment, a 3 kg water rocket is launched from the ground. The rocket's total energy at the top of its flight is 2352 J. (a) What was its launching speed? (b) What height did it reach? (c) What are its kinetic and potential energies after 2.5 s of launch?

Given Data

$$m = 3\ \text{kg}, \qquad E = 2352\ \text{J}, \qquad t = 2.5\ \text{s}$$

Assume vertical flight, negligible air resistance, and no further thrust after launch. Take gravitational potential energy as zero at the ground. The printed energy is **2352 J**.

To Find

(a) Launching speed u; (b) maximum height H; (c) KE and PE at $t = 2.5$ s.

Formula Used and Main Idea

Mechanical energy remains constant:

$$KE + PE = 2352\ \text{J}, \qquad KE = \frac{1}{2}mv^2, \qquad PE = mgh.$$

(a) Launching speed

At the ground, $PE = 0$, so all the mechanical energy is kinetic energy:

$$E = \frac{1}{2}mu^2.$$

Rearranging and substituting:

$$u^2 = \frac{2E}{m}, \qquad u = \sqrt{\frac{2(2352)}{3}} = \sqrt{1568}.$$

$$\boxed{u = 39.6\ \text{m s}^{-1}\ \text{upward}}$$

(b) Maximum height

At the highest point, the vertical speed is zero. All the energy is gravitational potential energy:

$$E = mgH.$$

$$H = \frac{E}{mg} = \frac{2352}{3(9.8)}.$$

$$\boxed{H = 80.0\ \text{m}}$$

Question 1 - continued

(c) Energies after 2.5 s

Choose upward as positive, so $a = -9.8\ \text{m s}^{-2}$. First find the speed:

$$v = u + at = 39.59798 - 9.8(2.5) = 15.09798\ \text{m s}^{-1}.$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}(3)(15.09798)^2.$$

$$\boxed{KE \approx 342\ \text{J}}$$

Now find the height at that instant:

$$y = ut + \frac{1}{2}at^2.$$

$$y = 39.59798(2.5) - \frac{1}{2}(9.8)(2.5)^2.$$

$$y = 98.99495 - 30.625 = 68.36995\ \text{m}.$$

$$PE = mgy = 3(9.8)(68.36995).$$

$$\boxed{PE \approx 2010\ \text{J}}$$

Check: $342 + 2010 = 2352$ J.

For unseen questions: At the highest point of a vertical flight, $v = 0$, but acceleration is still downward. Energy conservation applies to the assumed unpowered, constant-mass flight.

2. Which sphere requires more torque?

Question. Two spheres A and B have the same mass, but the radius of A is greater than that of B. Which one requires more torque to bring it into rotation about its axis?

Given Data

$$m_A = m_B, \qquad r_A > r_B.$$

To Find

Which sphere requires more torque for the **same angular acceleration**?

Formula Used

$$\tau = I\alpha.$$

For a uniform solid sphere rotating about a diameter:

$$I = \frac{2}{5}mr^2.$$

Solution

For equal masses:

$$I \propto r^2.$$

Since $r_A > r_B$:

$$I_A > I_B.$$

For the same angular acceleration α:

$$\tau_A = I_A\alpha > I_B\alpha = \tau_B.$$

Sphere A requires more torque.

Its mass is spread farther from the axis, so it has greater rotational inertia.

Assumption: The spheres have the same type of mass distribution and rotate about corresponding central axes.

3. Work done in moving a charge

Question. If 280 J of work is done in carrying a charge from a place where the potential is -12 V to another place where the potential is V volts, calculate V.

Given Data

$$W = 280\ \text{J}, \qquad V_i = -12\ \text{V}.$$

To Find

Final potential V_f.

Missing data: The question does not state the amount of charge q. Therefore, a unique numerical answer cannot be calculated.

Formula Used

For external work in moving a charge without changing its kinetic energy:

$$W = q(V_f - V_i).$$

Solution

$$280 = q[V_f - (-12)].$$

$$280 = q(V_f + 12).$$

Divide by q:

$$V_f + 12 = \frac{280}{q}.$$

$$\boxed{V_f = -12\ \text{V} + \frac{280\ \text{J}}{q}}$$

When q is in coulombs, $280\ \text{J}/q$ is in volts.

For unseen questions: $W = q\Delta V$. The charge must be known to find the potential difference from work. The equation above treats the stated work as external work; work done by the electric field has the opposite sign.

4. Number of beats in 10 seconds

Question. Two tuning forks of frequencies 456 Hz and 452 Hz are sounded together. How many beats are produced in 10 seconds?

Given Data

$$f_1 = 456 \text{ Hz}, \quad f_2 = 452 \text{ Hz}, \quad t = 10 \text{ s}.$$

To Find

Number of beats N.

Formula Used

$$f_b = |f_1 - f_2|, \qquad N = f_b t.$$

Solution

$$f_b = |456 - 452| = 4 \text{ Hz}.$$

This means four beats are produced each second.

$$N = 4(10).$$

$$\boxed{N = 40 \text{ beats}}$$

For unseen questions: Beat frequency gives beats per second. Multiply it by time to obtain the total number of beats.

5. Equal elongation of copper and aluminium wires

Question. A 40 kg mass is supported by a 5 m long aluminium wire with $Y_A = 7 \times 10^{10}$ Pa. To obtain the same elongation in a copper wire of the same length under the same weight, find the diameter of the copper wire. Take $Y_C = 12 \times 10^{10}$ Pa.

Given Data

$$m = 40 \text{ kg}, \qquad L_A = L_C = 5 \text{ m}.$$
$$Y_A = 7 \times 10^{10} \text{ Pa}, \qquad Y_C = 12 \times 10^{10} \text{ Pa}.$$

Both wires support the same force and have equal elongation.

To Find

Copper diameter d_C.

Missing data: The aluminium wire's diameter or area is not given. We can find only the diameter ratio.

Formula Used

$$Y = \frac{FL}{A\Delta L}, \qquad \Delta L = \frac{FL}{AY}.$$

Solution

For equal elongations:

$$\frac{FL}{A_A Y_A} = \frac{FL}{A_C Y_C}.$$

Cancel the common force and length:

$$A_A Y_A = A_C Y_C.$$
$$\frac{A_C}{A_A} = \frac{Y_A}{Y_C} = \frac{7}{12}.$$

Since $A = \pi d^2/4$:

$$\frac{d_C^2}{d_A^2} = \frac{7}{12}.$$

Taking square roots:

$$\frac{d_C}{d_A} = \sqrt{\frac{7}{12}} = 0.7638.$$
$$\boxed{d_C = 0.764 d_A}$$

The copper diameter should be about **76.4% of the aluminium diameter**.

Understanding: Copper has the larger Young modulus, so a thinner copper wire can produce the same extension under the same load and length.

6. Projection angle when height equals range

Question. Find the angle of projection of a projectile for which its maximum height and corresponding horizontal range are equal.

Given Data

$$H = R.$$

Assume the projectile lands at the launch level and air resistance is negligible.

To Find

Angle of projection θ.

Formula Used

$$H = \frac{u^2 \sin^2\theta}{2g}, \qquad R = \frac{u^2 \sin 2\theta}{g}.$$

Solution

Set $H = R$:

$$\frac{u^2 \sin^2\theta}{2g} = \frac{u^2 \sin 2\theta}{g}.$$

Cancel u^2/g:

$$\frac{\sin^2\theta}{2} = \sin 2\theta.$$

Use $\sin 2\theta = 2\sin\theta\cos\theta$:

$$\sin^2\theta = 4\sin\theta\cos\theta.$$

For a nonzero trajectory, divide by $\sin\theta$:

$$\sin\theta = 4\cos\theta.$$

$$\tan\theta = 4.$$

$$\theta = \tan^{-1}(4).$$

$$\boxed{\theta \approx 76.0^\circ}$$

For unseen questions: Write the two projectile formulas first, apply the given relationship, and cancel common quantities before using trigonometric identities.

7. Accuracy of surface area and volume

Question. The radius of a sphere is measured using a screw gauge and substituted into the formulas for its surface area (printed as $A = \pi r^2$) and volume $V = \frac{4}{3}\pi r^3$. Which result is more accurate?

Given Data

The same measured radius r, with uncertainty Δr, is used for both calculations.

To Find

Which result has the smaller percentage uncertainty?

Correction: The surface area of a sphere is $A = 4\pi r^2$. The printed $A = \pi r^2$ is not the sphere's surface area. This correction does not change the uncertainty comparison.

Formula Used

If $Q \propto r^n$, the approximate fractional uncertainty is:

$$\frac{\Delta Q}{Q} = n\frac{\Delta r}{r}.$$

Solution

For surface area:

$$A = 4\pi r^2, \qquad \frac{\Delta A}{A} = 2\frac{\Delta r}{r}.$$

For volume:

$$V = \frac{4}{3}\pi r^3, \qquad \frac{\Delta V}{V} = 3\frac{\Delta r}{r}.$$

Therefore:

$$\frac{\Delta A}{A} < \frac{\Delta V}{V}.$$

The surface-area result is more accurate.

For example, a 1% radius uncertainty gives approximately 2% area uncertainty and 3% volume uncertainty.

Understanding: Raising a measured quantity to a higher power increases its fractional uncertainty.

8. KE at the top equals one-fourth of initial KE

Question. Calculate the angle of projection for which the kinetic energy at the highest point of the trajectory is one-fourth of the kinetic energy at the point of projection.

Given Data

$$KE_{\text{top}} = \frac{1}{4} KE_i.$$

To Find

Projection angle θ.

Formula Used

$$KE = \frac{1}{2} mv^2.$$

At the highest point, $v_y = 0$ but the horizontal component remains:

$$v_x = u \cos\theta.$$

Solution

Initially:

$$KE_i = \frac{1}{2} mu^2.$$

At the top:

$$KE_{\text{top}} = \frac{1}{2} m(u \cos\theta)^2.$$

Apply the condition:

$$\frac{1}{2} mu^2 \cos^2\theta = \frac{1}{4} \left(\frac{1}{2} mu^2 \right).$$

Cancel $\frac{1}{2} mu^2$:

$$\cos^2\theta = \frac{1}{4}.$$

For an upward projection between 0° and 90°:

$$\cos\theta = \frac{1}{2}.$$

$$\boxed{\theta = 60^\circ}$$

For unseen questions: At the top of an oblique projectile's path, only its vertical velocity becomes zero. Its total kinetic energy is generally not zero.

9. Water speed in connected pipes

Question. Water flows at $5\ \text{ft}\,\text{s}^{-1}$ in a pipe of diameter 6 inches. Find its speed in a connected pipe of diameter 3 inches.

Given Data

$$d_1 = 6\ \text{in}, \quad d_2 = 3\ \text{in}, \quad v_1 = 5\ \text{ft}\,\text{s}^{-1}.$$

To Find

Speed v_2 in the smaller pipe.

Formula Used

For steady incompressible flow:

$$A_1 v_1 = A_2 v_2.$$

Since $A = \pi d^2/4$:

$$v_2 = \left(\frac{d_1}{d_2}\right)^2 v_1.$$

Solution

Both diameters have the same unit, so their units cancel in the ratio:

$$v_2 = \left(\frac{6}{3}\right)^2 (5) = 4(5).$$

$$\boxed{v_2 = 20\ \text{ft}\,\text{s}^{-1}}$$

In SI units, using $1\ \text{ft} = 0.3048\ \text{m}$:

$$v_2 = 20(0.3048).$$

$$\boxed{v_2 = 6.096\ \text{m}\,\text{s}^{-1}}$$

Understanding: Halving the diameter reduces area to one-fourth, so the speed becomes four times greater.

10. Frequency heard from a receding bus

Question. A bus horn emits sound of frequency 200 Hz. The bus moves away from a stationary observer at 18 m/s. If the speed of sound is 331 m/s, what frequency does the observer hear?

Given Data

$$f_s = 200\ \text{Hz}, \quad v_s = 18\ \text{m}\,\text{s}^{-1}, \quad v = 331\ \text{m}\,\text{s}^{-1}.$$

The observer is stationary.

To Find

Observed frequency f_o.

Formula Used

For a source moving away from a stationary observer:

$$f_o = f_s \frac{v}{v + v_s}.$$

Solution

$$f_o = 200 \left(\frac{331}{331 + 18} \right).$$

$$f_o = \frac{66200}{349}.$$

$$\boxed{f_o \approx 189.7\ \text{Hz}}$$

Check: A receding source produces a lower observed frequency, so the result must be below 200 Hz.

11. Maximum height of a vertically thrown ball

Question. A ball of mass 100 g is thrown vertically upward at 25 m/s. If no energy is lost, determine the height it reaches.

Given Data

$$m = 100\ \text{g} = 0.100\ \text{kg}, \quad u = 25\ \text{m s}^{-1}.$$

At the highest point, $v = 0$.

To Find

Maximum height H above the point of projection.

Formula Used

With no energy loss:

$$\frac{1}{2}mu^2 = mgH.$$

Solution

Cancel m:

$$H = \frac{u^2}{2g}.$$

$$H = \frac{25^2}{2(9.8)} = \frac{625}{19.6}.$$

$$\boxed{H \approx 31.9\ \text{m}}$$

For unseen questions: Mass cancels. In this ideal situation, maximum height depends on launching speed and gravity, not on mass.

12. Dimensional correctness of the velocity equation

Question. Prove that the equation $v_f = v_i + at$ is dimensionally correct. The source appears to print f_i in place of v_i.

Given Data

$$v_f = v_i + at.$$

Here v_f is final velocity, v_i is initial velocity, a is acceleration, and t is time.

To Show

Every term in the equation has the same dimensions.

Dimensions Used

$$[v_f] = [v_i] = LT^{-1}, \qquad [a] = LT^{-2}, \qquad [t] = T.$$

Solution

The dimensions of the product at are:

$$[at] = [a][t] = (LT^{-2})(T) = LT^{-1}.$$

Therefore:

$$[v_f] = [v_i] = [at] = LT^{-1}.$$

The equation is dimensionally correct.

Only quantities with the same dimensions can be added. If the printed symbol f_i were actually intended to mean frequency, the equation would not be dimensionally correct.

13. Work done on a suitcase

Question. A girl pulls a suitcase 4.90 m along level ground with a constant force of 16.8 N at 30° above the horizontal. A frictional force of 0.170 N opposes its motion. Find (a) work done by the girl, (b) work done by friction, and (c) net work done by all forces.

Given Data

$$s = 4.90\ \text{m}, \quad F = 16.8\ \text{N}, \quad \theta = 30^\circ, \quad f = 0.170\ \text{N}.$$

To Find

W_{girl}, W_f, and W_{net}.

Formula Used

$$W = Fs\cos\theta.$$

Here θ is the angle **between force and displacement**.

(a) Work done by the girl

$$W_{\text{girl}} = 16.8(4.90)\cos 30^\circ.$$

$$W_{\text{girl}} = 16.8(4.90)(0.8660).$$

$$\boxed{W_{\text{girl}} \approx 71.3\ \text{J}}$$

(b) Work done by friction

Friction acts opposite to the displacement, so its angle is 180°:

$$W_f = fs\cos 180^\circ.$$

$$W_f = 0.170(4.90)(-1).$$

$$\boxed{W_f = -0.833\ \text{J}}$$

(c) Net work

Weight and normal reaction are perpendicular to horizontal displacement:

$$W_{\text{weight}} = W_{\text{normal}} = 0.$$

Therefore:

$$W_{\text{net}} = 71.2912 - 0.833.$$

$$\boxed{W_{\text{net}} \approx 70.5\ \text{J}}$$

Understanding: Positive work adds kinetic energy; negative work removes it. A force perpendicular to displacement does zero work.

14. Electric force on a charge

Question. An electric field has a strength of 5 N/C. What force acts on a charge of 2 C placed in this field?

Given Data

$$E = 5\,\mathrm{N\,C^{-1}}, \qquad q = 2\,\mathrm{C}.$$

To Find

Electric force F.

Formula Used

$$E = \frac{F}{q} \quad \Rightarrow \quad F = qE.$$

Solution

$$F = 2(5).$$

$$\boxed{F = 10\,\mathrm{N}}$$

Since the charge is positive, the force is **along the electric field**.

Unit check: $\mathrm{C} \times \mathrm{N\,C^{-1}} = \mathrm{N}$.

15. Strain energy: extension 0.0039 m

Question. Calculate the strain energy stored in a 1 m long steel wire suspended from the roof and supporting a weight of 980 N. Its diameter is 0.004 m and its extension is $\Delta L = 0.0039$ m.

Given Data

$$L = 1 \text{ m}, \quad F = 980 \text{ N}, \quad d = 0.004 \text{ m}.$$
$$\Delta L = 0.0039 \text{ m}.$$

To Find

Strain energy U.

Formula Used

For a linearly elastic wire loaded gradually from zero:

$$U = \frac{1}{2} F \Delta L.$$

Solution

$$U = \frac{1}{2}(980)(0.0039).$$
$$U = 490(0.0039).$$
$$\boxed{U = 1.911 \text{ J} \approx 1.91 \text{ J}}$$

Why the factor $\frac{1}{2}$? The stretching force rises from zero to 980 N, so its average value is 490 N. The stored energy equals the area under the straight-line force-extension graph.

The length and diameter are unnecessary here because the final force and extension are already given.

16. Coulomb's law and separation of charges

Question. State Coulomb's law and write its mathematical expression. Two point charges $q_1 = 4\ \mu$C and $q_2 = 2\ \mu$C repel each other with a force of 36 N. Calculate their separation.

Statement of Coulomb's Law

The electrostatic force between two stationary point charges is directly proportional to the product of their charge magnitudes and inversely proportional to the square of their separation. The force acts along the line joining them.

$$F = k\frac{|q_1 q_2|}{r^2}.$$

Like charges repel; unlike charges attract.

Given Data

$$q_1 = 4\ \mu\text{C} = 4 \times 10^{-6}\ \text{C}.$$

$$q_2 = 2\ \mu\text{C} = 2 \times 10^{-6}\ \text{C}, \qquad F = 36\ \text{N}.$$

To Find

Separation r.

Solution

Rearrange Coulomb's law:

$$r^2 = \frac{k|q_1 q_2|}{F}.$$

$$r = \sqrt{\frac{(9 \times 10^9)(4 \times 10^{-6})(2 \times 10^{-6})}{36}}.$$

$$r = \sqrt{\frac{0.072}{36}} = \sqrt{0.002}.$$

$$\boxed{r = 0.0447\ \text{m} = 4.47\ \text{cm}}$$

For unseen questions: Convert microcoulombs to coulombs before substitution. Remember to take the square root when finding distance.

17. Components of the trolley-pushing force

Question. A peddler pushes a trolley along a horizontal road with a force of 50 N making an angle of 30° with the road. Find the horizontal and vertical components of the force.

Given Data

$$F = 50\ \text{N}, \qquad \theta = 30^\circ.$$

To Find

Horizontal component F_x and vertical component F_y.

Formula Used

$$F_x = F\cos\theta, \qquad |F_y| = F\sin\theta.$$

Solution

Horizontal component:

$$F_x = 50\cos 30^\circ = 50(0.8660).$$

$$\boxed{F_x = 43.3\ \text{N forward}}$$

Vertical component magnitude:

$$|F_y| = 50\sin 30^\circ = 50(0.5).$$

$$\boxed{|F_y| = 25.0\ \text{N}}$$

For the usual downward push on a trolley, with upward taken as positive:

$$\boxed{F_y = -25.0\ \text{N}}.$$

If the force were directed above the horizontal, this component would instead be upward.

For unseen questions: When the angle is measured from the horizontal, the horizontal component uses cosine and the vertical component uses sine.

18. Resistance and uncertainty

Question. The measured voltage across a conductor is (7.3 ± 0.3) V and its current is (2.73 ± 0.051) A. Find the resistance and its uncertainty.

Given Data

$$V = (7.3 \pm 0.3)\ \text{V}, \qquad I = (2.73 \pm 0.051)\ \text{A}.$$

To Find

Resistance R and absolute uncertainty ΔR.

Formula Used

$$R = \frac{V}{I}.$$

Using the usual school-level maximum-uncertainty rule:

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}.$$

Solution

First calculate resistance:

$$R = \frac{7.3}{2.73} = 2.67399\ \Omega.$$

Now calculate fractional uncertainty:

$$\frac{\Delta R}{R} = \frac{0.3}{7.3} + \frac{0.051}{2.73}.$$

$$\frac{\Delta R}{R} = 0.041096 + 0.018681 = 0.059777.$$

Percentage uncertainty:

$$0.059777(100) \approx 5.98\%.$$

Absolute uncertainty:

$$\Delta R = R(0.059777).$$

$$\Delta R = 2.67399(0.059777) = 0.15984\ \Omega.$$

$$\boxed{R = (2.67 \pm 0.16)\ \Omega}$$

For unseen questions: In multiplication or division, add fractional uncertainties using this exam convention. Finally convert fractional uncertainty back to absolute uncertainty.

19. Proton mass tripled

Question. At what speed would the mass of a proton be tripled? Its rest mass is 1.673×10^{-27} kg.

Given Data

$$m = 3m_0, \qquad m_0 = 1.673 \times 10^{-27}\ \text{kg}.$$
$$c = 3 \times 10^{8}\ \text{m s}^{-1}.$$

To Find

Speed v.

Formula Used

Using the relativistic-mass convention in the question and formula sheet:

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}.$$

Solution

$$3m_0 = \frac{m_0}{\sqrt{1 - v^2/c^2}}.$$

Cancel m_0:

$$3 = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

Take reciprocals:

$$\sqrt{1 - v^2/c^2} = \frac{1}{3}.$$

Square both sides:

$$1 - \frac{v^2}{c^2} = \frac{1}{9}.$$
$$\frac{v^2}{c^2} = \frac{8}{9}.$$
$$v = c\sqrt{\frac{8}{9}} = (3 \times 10^{8})(0.942809).$$
$$\boxed{v = 2.83 \times 10^{8}\ \text{m s}^{-1} \approx 0.943c}$$

The rest mass cancels because only the ratio m/m_0 is needed.

20. Magnetic field strength

Question. A particle of charge $-3e$ moves at 1×10^6 m/s at an angle of 45° to a magnetic field. It experiences a magnetic force of 6.72×10^{-14} N. Find the field strength.

Given Data

$$q = -3e, \qquad |q| = 3(1.6 \times 10^{-19}) = 4.8 \times 10^{-19}\ \text{C}.$$
$$v = 1 \times 10^6\ \text{m s}^{-1}, \quad \theta = 45^\circ, \quad F = 6.72 \times 10^{-14}\ \text{N}.$$

To Find

Magnetic field strength B.

Formula Used

$$F = |q| v B \sin\theta.$$

Solution

Rearrange:

$$B = \frac{F}{|q| v \sin\theta}.$$
$$B = \frac{6.72 \times 10^{-14}}{(4.8 \times 10^{-19})(1 \times 10^6)\sin 45^\circ}.$$
$$B = \frac{6.72 \times 10^{-14}}{3.3941 \times 10^{-13}}.$$
$$\boxed{B \approx 0.198\ \text{T}}$$

Understanding: The negative charge affects force direction. Use the charge magnitude when calculating the magnetic field strength.

21. Strain energy: extension 0.00039 m

Question. Calculate the strain energy stored in a 1 m long steel wire suspended from the roof and supporting a weight of 980 N. Its diameter is 0.004 m and its extension is $\Delta L = 0.00039$ m.

Given Data

$$L = 1 \text{ m}, \quad F = 980 \text{ N}, \quad d = 0.004 \text{ m}.$$
$$\Delta L = 0.00039 \text{ m}.$$

To Find

Strain energy U.

Formula Used

For a linearly elastic wire loaded gradually:

$$U = \frac{1}{2} F \Delta L.$$

Solution

$$U = \frac{1}{2}(980)(0.00039).$$
$$U = 490(0.00039).$$
$$\boxed{U = 0.1911 \text{ J} \approx 0.191 \text{ J}}$$

Comparison with Question 15: This extension is ten times smaller, so the energy is ten times smaller for the same final force. Both printed versions have been solved separately.

The length and diameter are not required because force and extension are known.

22. Vibrating string: frequencies and wave speed

Question. A string has a frequency of 125 Hz when it vibrates in five segments. Find its frequencies when vibrating in three and four segments. Also find the wave speed if the string length is 0.80 m.

Given Data

$$f_5 = 125 \text{ Hz}, \qquad L = 0.80 \text{ m}.$$

To Find

f_3, f_4, and wave speed v.

Formula Used

For a string fixed at both ends, at unchanged tension and linear density:

$$f_n = nf_1, \qquad L = \frac{n\lambda_n}{2}, \qquad v = f_n\lambda_n.$$

Solution

Five segments correspond to the fifth harmonic:

$$f_5 = 5f_1.$$

$$f_1 = \frac{125}{5} = 25 \text{ Hz}.$$

For three segments:

$$f_3 = 3f_1 = 3(25).$$

$$\boxed{f_3 = 75 \text{ Hz}}$$

For four segments:

$$f_4 = 4f_1 = 4(25).$$

$$\boxed{f_4 = 100 \text{ Hz}}$$

For the fifth harmonic:

$$\lambda_5 = \frac{2L}{5} = \frac{2(0.80)}{5} = 0.32 \text{ m}.$$

$$v = f_5\lambda_5 = 125(0.32).$$

$$\boxed{v = 40 \text{ m s}^{-1}}$$

For unseen questions: Each segment is one half-wavelength. Thus n segments fit n half-wavelengths into the string length.

23. Resistance of a copper wire

Question. A copper wire is 50 m long and has a cross-sectional area of 1.5×10^{-6} m^2. Its resistivity is 1.68×10^{-8} Ωm. Find its resistance.

Given Data

$$L = 50 \text{ m}, \qquad A = 1.5 \times 10^{-6} \text{ m}^2.$$
$$\rho = 1.68 \times 10^{-8} \ \Omega\text{m}.$$

To Find

Resistance R.

Formula Used

$$R = \rho \frac{L}{A}.$$

Solution

$$R = \frac{(1.68 \times 10^{-8})(50)}{1.5 \times 10^{-6}}.$$
$$R = \frac{8.4 \times 10^{-7}}{1.5 \times 10^{-6}}.$$
$$\boxed{R = 0.56 \ \Omega}$$

For unseen questions: Resistance increases with length and decreases with cross-sectional area.

24. Angle between two vectors

Question. If $|\vec{A} \times \vec{B}| = 3/4$ and $|\vec{A} \cdot \vec{B}| = 1/2$, find the angle between $\vec{A}$ and $\vec{B}$.

Given Data

$$|\vec{A} \times \vec{B}| = \frac{3}{4}, \qquad |\vec{A} \cdot \vec{B}| = \frac{1}{2}.$$

To Find

Angle θ.

Formula Used

$$|\vec{A} \times \vec{B}| = AB \sin\theta.$$
$$|\vec{A} \cdot \vec{B}| = AB|\cos\theta|.$$

Solution

Divide the two expressions:

$$\frac{AB \sin\theta}{AB|\cos\theta|} = \frac{3/4}{1/2}.$$
$$|\tan\theta| = \frac{3}{2}.$$

The acute solution is:

$$\theta_1 = \tan^{-1}(1.5).$$
$$\boxed{\theta_1 = 56.3^\circ}$$

However, the question gives the **absolute value** of the dot product. Therefore, an obtuse angle also satisfies the data:

$$\theta_2 = 180^\circ - 56.3^\circ.$$
$$\boxed{\theta_2 = 123.7^\circ}$$

Expected answer if a positive dot product or an acute angle is intended: 56.3°.

Strictly, the printed absolute-value information permits both angles. Without the sign of the dot product, they cannot be distinguished.

25. Resistance from length and diameter

Question. Calculate the resistance of a 15 m long wire with a diameter of 1.5 mm and resistivity $2.63 \times 10^{-2}\ \Omega$m.

Given Data

$$L = 15\ \text{m}, \qquad d = 1.5\ \text{mm} = 1.5 \times 10^{-3}\ \text{m}.$$
$$\rho = 2.63 \times 10^{-2}\ \Omega\text{m}.$$

To Find

Resistance R.

Formula Used

$$R = \rho \frac{L}{A}, \qquad A = \pi r^2.$$

Solution

First find the radius:

$$r = \frac{d}{2} = \frac{1.5 \times 10^{-3}}{2} = 7.5 \times 10^{-4}\ \text{m}.$$

Then find area:

$$A = \pi (7.5 \times 10^{-4})^2.$$
$$A = 1.7671 \times 10^{-6}\ \text{m}^2.$$

Substitute into the resistance formula:

$$R = \frac{(2.63 \times 10^{-2})(15)}{1.7671 \times 10^{-6}}.$$
$$R = \frac{0.3945}{1.7671 \times 10^{-6}}.$$
$$\boxed{R \approx 2.23 \times 10^{5}\ \Omega}$$

Printing note: The supplied question shows 10^{-2}, which gives this large resistance. If 10^{-8} was intended instead, the result would be $0.223\ \Omega$. Use the exponent actually supplied in the exam.

Common mistake: Use $r = d/2$ in $A = \pi r^2$, or directly use $A = \pi d^2/4$.

26. Proton moving in a circular magnetic orbit

Question. A proton moves in a circular orbit of radius 12 mm in a uniform 0.30 T magnetic field perpendicular to its velocity. Find its speed.

Given Data

$$r = 12\ \text{mm} = 0.012\ \text{m}, \qquad B = 0.30\ \text{T}.$$
$$q = 1.6 \times 10^{-19}\ \text{C}, \qquad m_p = 1.673 \times 10^{-27}\ \text{kg}.$$

The velocity is perpendicular to the field, so $\sin 90^\circ = 1$.

To Find

Speed v.

Formula Used

Magnetic force supplies the centripetal force:

$$qvB = \frac{mv^2}{r}.$$

Solution

Cancel one factor of v:

$$qB = \frac{mv}{r}.$$

Rearrange:

$$v = \frac{qBr}{m}.$$
$$v = \frac{(1.6 \times 10^{-19})(0.30)(0.012)}{1.673 \times 10^{-27}}.$$
$$v = \frac{5.76 \times 10^{-22}}{1.673 \times 10^{-27}}.$$
$$\boxed{v \approx 3.44 \times 10^{5}\ \text{m s}^{-1}}$$

Understanding: The magnetic force changes the direction of velocity and maintains circular motion.

27. Currents in the five-resistor circuit

Question. Find the current through each resistor in the given circuit using Kirchhoff's laws.

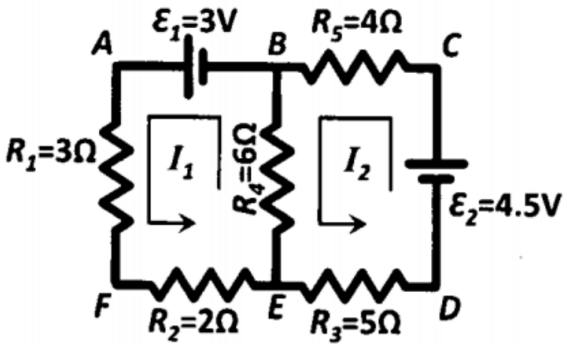


Figure 1: Circuit supplied in the question.

Given Data

$$R_1 = 3\,\Omega, \quad R_2 = 2\,\Omega, \quad R_3 = 5\,\Omega, \quad R_4 = 6\,\Omega, \quad R_5 = 4\,\Omega.$$
$$\varepsilon_1 = 3\ \text{V}, \qquad \varepsilon_2 = 4.5\ \text{V}.$$

The long battery plate is the positive terminal.

To Find

Current through every resistor, including its direction.

Laws Used

Kirchhoff's junction law:

$$\sum I_{\text{in}} = \sum I_{\text{out}}.$$

Kirchhoff's loop law:

$$\sum \varepsilon - \sum IR = 0.$$

Step 1: Choose loop currents

Choose I_1 around $A \to F \to E \to B \to A$, and I_2 around $B \to E \to D \to C \to B$.

In the shared resistor R_4, the loop currents oppose each other. Taking $E \to B$ as positive:

$$I_{R_4} = I_1 - I_2.$$

Step 2: Left-loop equation

The 3 V cell is crossed from negative to positive:

$$3 - 3I_1 - 2I_1 - 6(I_1 - I_2) = 0.$$

Expand:

$$3 - 3I_1 - 2I_1 - 6I_1 + 6I_2 = 0.$$

$$\boxed{11I_1 - 6I_2 = 3} \tag{1}$$

Question 27 - continued

Step 3: Right-loop equation

The 4.5 V cell is crossed from negative to positive:

$$4.5 - 5I_2 - 4I_2 - 6(I_2 - I_1) = 0.$$

$$\boxed{-6I_1 + 15I_2 = 4.5} \tag{2}$$

Step 4: Solve simultaneously

Multiply Equation (1) by 15:

$$165I_1 - 90I_2 = 45.$$

Multiply Equation (2) by 6:

$$-36I_1 + 90I_2 = 27.$$

Add the equations:

$$129I_1 = 72.$$

$$I_1 = \frac{72}{129} = 0.55814 \text{ A}.$$

Substitute into Equation (2):

$$-6(0.55814) + 15I_2 = 4.5.$$

$$15I_2 = 7.84884, \qquad I_2 = 0.52326 \text{ A}.$$

Shared current:

$$I_{R_4} = 0.55814 - 0.52326 = 0.03488 \text{ A}.$$

Final Answers

Resistor	Current	Direction
$R_1 = 3\,\Omega$	0.558 A	$A \to F$
$R_2 = 2\,\Omega$	0.558 A	$F \to E$
$R_3 = 5\,\Omega$	0.523 A	$E \to D$
$R_4 = 6\,\Omega$	0.0349 A	$E \to B$
$R_5 = 4\,\Omega$	0.523 A	$C \to B$

Check at E: $0.55814 = 0.52326 + 0.03488$ A.

For unseen questions: In a shared resistor, use the difference of opposing loop currents. A negative calculated current means the actual direction is opposite to the assumed direction.

28. Charged object floating in an electric field

Question. An object with a net charge of $24\ \mu$C is placed in a uniform electric field of 610 N/C directed vertically. What is its mass if it floats in the field?

Given Data

$$q = 24\ \mu\text{C} = 24 \times 10^{-6}\ \text{C}.$$
$$E = 610\ \text{N C}^{-1}.$$

To Find

Mass m.

Formula Used

For the object to float, electric force balances weight:

$$F_e = F_g, \qquad qE = mg.$$

Solution

$$m = \frac{qE}{g}.$$
$$m = \frac{(24 \times 10^{-6})(610)}{9.8}.$$
$$m = \frac{0.01464}{9.8}.$$
$$\boxed{m = 1.49 \times 10^{-3}\ \text{kg} = 1.49\ \text{g}}$$

Direction: Because the charge is positive, the electric field must point upward to provide an upward electric force against the downward weight.

29. Distance using electric field and potential

Question. The electric field at a point due to a point charge is 26 N/C, and the electric potential there is 13 J/C. Find the distance of the point from the charge.

Given Data

$$E = 26\ \mathrm{N\,C^{-1}}, \qquad V = 13\ \mathrm{J\,C^{-1}} = 13\ \mathrm{V}.$$

To Find

Distance r.

Formula Used

For a point charge, with potential zero at infinity:

$$E = \frac{kQ}{r^2}, \qquad V = \frac{kQ}{r}.$$

Using magnitudes and dividing the expressions gives:

$$\frac{V}{E} = r, \qquad V = Er.$$

Solution

$$r = \frac{V}{E} = \frac{13}{26}.$$

$$\boxed{r = 0.50\ \mathrm{m}}$$

For unseen questions: $V = Er$ here follows specifically from the point-charge formulas. It is not a general formula for every electric-field arrangement.

30. Falling ball with air resistance

Question. A 120 g ball is released from a height of 30 m. If it encounters an air resistance of 0.3 N, find its kinetic energy just before striking the ground.

Given Data

$$m = 120\text{ g} = 0.120\text{ kg}, \quad h = 30\text{ m}, \quad f = 0.3\text{ N}.$$

The ball is released from rest. Treat the air resistance as constant.

To Find

Final kinetic energy KE_f.

Formula Used

$$KE_f = mgh - fh.$$

Here fh is the positive amount of energy lost against resistance.

Solution

Initial gravitational potential energy:

$$PE_i = mgh = 0.120(9.8)(30) = 35.28\text{ J}.$$

Energy lost against air resistance:

$$E_{\text{lost}} = fh = 0.3(30) = 9.0\text{ J}.$$

Therefore:

$$KE_f = 35.28 - 9.0.$$

$$\boxed{KE_f = 26.28\text{ J} \approx 26.3\text{ J}}$$

Formula-sheet clarification: Signed work done by resistance is negative: $W_{\text{resistance}} = -fh$. Therefore, the equivalent signed-work expression is $KE_f = mgh + W_{\text{resistance}}$.

For unseen questions: First calculate available gravitational energy, then subtract the energy dissipated by resistance.

31. Electric field of a point charge

Question. Calculate the electric field intensity at a distance of 0.5 m from a point charge of 2×10^{-6} C in a vacuum.

Given Data

$$Q = 2 \times 10^{-6}\ \text{C}, \qquad r = 0.5\ \text{m}.$$

To Find

Electric field intensity E.

Formula Used

$$E = \frac{k|Q|}{r^2}.$$

Solution

$$E = \frac{(9 \times 10^9)(2 \times 10^{-6})}{(0.5)^2}.$$

$$E = \frac{18 \times 10^3}{0.25}.$$

$$\boxed{E = 7.2 \times 10^4\ \text{N C}^{-1}}$$

The field points **radially away from the positive charge**.

For unseen questions: The distance is squared. Doubling the distance reduces the field magnitude to one-fourth.

32. Heat-engine efficiency

Question. Find the efficiency of a heat engine operating between 427°C and 27°C.

Given Data

Convert both temperatures to kelvin:

$$T_H = 427 + 273 = 700\ \text{K}.$$

$$T_C = 27 + 273 = 300\ \text{K}.$$

To Find

Maximum theoretical efficiency η.

Formula Used

For a reversible Carnot engine:

$$\eta = 1 - \frac{T_C}{T_H}.$$

Solution

$$\eta = 1 - \frac{300}{700}.$$

$$\eta = 1 - 0.42857 = 0.57143.$$

Convert to percentage:

$$\eta\% = 0.57143(100).$$

$$\boxed{\eta_{\text{max}} \approx 57.1\%}$$

Important: Temperatures alone determine the Carnot maximum. A real engine's actual efficiency cannot be found without more information and cannot exceed this maximum.

Common mistake: Do not substitute Celsius temperatures into the temperature ratio.

33. Electron drift velocity

Question. A current of 2 A flows through a copper conductor of cross-sectional area 1 mm^2. Find the drift velocity of electrons. For copper, $n = 8.5 \times 10^{28}$ electrons per m^3.

Given Data

$$I = 2\ \text{A}, \qquad A = 1\ \text{mm}^2 = 1 \times 10^{-6}\ \text{m}^2.$$
$$n = 8.5 \times 10^{28}\ \text{m}^{-3}, \qquad e = 1.6 \times 10^{-19}\ \text{C}.$$

To Find

Drift speed v_d and its direction relative to conventional current.

Formula Used

$$I = nAev_d.$$

Solution

$$v_d = \frac{I}{nAe}.$$
$$v_d = \frac{2}{(8.5 \times 10^{28})(1 \times 10^{-6})(1.6 \times 10^{-19})}.$$

Combine numbers and powers:

$$nAe = 13.6 \times 10^3 = 13600.$$
$$v_d = \frac{2}{13600}.$$
$$\boxed{v_d = 1.47 \times 10^{-4}\ \text{m s}^{-1}}$$

Electron drift is **opposite to conventional current**.

Conversion reminder:

$$1\ \text{mm}^2 = (10^{-3}\ \text{m})^2 = 10^{-6}\ \text{m}^2.$$

34. Sound intensity at a distance

Question. A point source emits sound energy at a rate of 15 W. Calculate the sound intensity 4 m from the source.

Given Data

$$P = 15\ \text{W}, \qquad r = 4\ \text{m}.$$

To Find

Sound intensity I.

Formula Used

Assuming uniform emission in all directions:

$$I = \frac{P}{A}, \qquad A = 4\pi r^2.$$

Thus:

$$I = \frac{P}{4\pi r^2}.$$

Solution

$$I = \frac{15}{4\pi(4)^2}.$$

$$I = \frac{15}{64\pi} = \frac{15}{201.06}.$$

$$\boxed{I = 0.0746\ \text{W m}^{-2}}$$

For unseen questions: At twice the distance, intensity becomes one-fourth, assuming negligible absorption. The relevant area is the surface area of a sphere, $4\pi r^2$.

35. Horizontal launch from a cliff

Question. A projectile is launched horizontally from a 50 m high cliff and lands 20 m from the cliff's base. Determine the magnitude of the launch velocity.

Given Data

$$h = 50\ \text{m}, \qquad x = 20\ \text{m}, \qquad u_y = 0.$$

Assume negligible air resistance.

To Find

Horizontal launch speed u_x.

Formula Used

Vertical motion, taking downward as positive:

$$h = u_y t + \frac{1}{2} g t^2.$$

Horizontal motion:

$$x = u_x t.$$

Step 1: Find flight time

Since $u_y = 0$:

$$50 = \frac{1}{2}(9.8)t^2.$$

$$t^2 = \frac{100}{9.8}.$$

$$t = 3.1944\ \text{s}.$$

Step 2: Find launch speed

$$u_x = \frac{x}{t} = \frac{20}{3.1944}.$$

$$\boxed{u_x = 6.26\ \text{m s}^{-1}}$$

Understanding: Vertical motion determines the flight time. Horizontal motion then gives the launching speed. Both motions occur over the same time interval.

36. Dimensions of Planck's constant

Question. The energy of a photon is $E = hf$, where f is frequency. Find the dimensions of Planck's constant h.

Given Data

$$E = hf.$$

To Find

Dimensions and SI unit of h.

Dimensions Used

$$[E] = ML^2T^{-2}, \qquad [f] = T^{-1}.$$

Solution

Rearrange:

$$h = \frac{E}{f}.$$

$$[h] = \frac{ML^2T^{-2}}{T^{-1}}.$$

$$\boxed{[h] = ML^2T^{-1}}$$

SI unit:

$$\boxed{\text{J s}}$$

Equivalent SI base units:

$$\boxed{\text{kg m}^2\,\text{s}^{-1}}$$

For unseen questions: Rearrange the physical equation first, then replace each quantity with its dimensions.

37. Power from force and velocity vectors

Question. Find the power delivered by an engine when its velocity is $(4\hat{i} + 5\hat{j})$ m/s and the force it exerts is $(8\hat{i} - 2\hat{j})$ N.

Given Data

$$\vec{v} = (4\hat{i} + 5\hat{j})\ \mathrm{m\,s^{-1}}.$$
$$\vec{F} = (8\hat{i} - 2\hat{j})\ \mathrm{N}.$$

To Find

Instantaneous power P.

Formula Used

$$P = \vec{F} \cdot \vec{v}.$$

In components:

$$P = F_x v_x + F_y v_y.$$

Solution

$$P = (8)(4) + (-2)(5).$$
$$P = 32 - 10.$$
$$\boxed{P = 22\ \mathrm{W}}$$

For unseen questions: Power uses the dot product. Multiply corresponding components and then add. Here the y component contributes negative power.

38. Rabbit jumping over a 0.7 m object

Question. A rabbit jumps over an object 0.7 m high. (a) Considering its vertical motion, find its vertical speed when leaving the ground. (b) How long does it remain in the air?

Given Data

$$H = 0.7 \text{ m}, \qquad v_y = 0 \text{ at the highest point.}$$

Assume 0.7 m is the maximum height of the jump and the rabbit lands at the launch level.

To Find

(a) Initial vertical speed u_y; (b) total flight time T.

(a) Initial vertical speed

Taking upward as positive, $a = -g$. Use:

$$v_y^2 = u_y^2 + 2aH.$$

At maximum height:

$$0 = u_y^2 - 2gH.$$

$$u_y = \sqrt{2gH}.$$

$$u_y = \sqrt{2(9.8)(0.7)} = \sqrt{13.72}.$$

$$\boxed{u_y = 3.70 \text{ m s}^{-1} \text{ upward}}$$

(b) Time in air

At the top:

$$v_y = u_y - gt_{\text{up}}.$$

$$0 = u_y - gt_{\text{up}}.$$

$$t_{\text{up}} = \frac{u_y}{g} = \frac{3.70405}{9.8} = 0.37796 \text{ s}.$$

For landing at the same level:

$$T = 2t_{\text{up}}.$$

$$\boxed{T = 0.756 \text{ s}}$$

Interpretation: If the rabbit rises higher than 0.7 m, these are the minimum values under the same-level landing assumption.

39. Power for velocity (2, 3)

Question. Find the power delivered by an engine when its velocity is $(2, 3)$ m/s and its force is $(7, -3)$ N.

Given Data

$$\vec{v} = (2\hat{i} + 3\hat{j})\ \mathrm{m\,s^{-1}}.$$
$$\vec{F} = (7\hat{i} - 3\hat{j})\ \mathrm{N}.$$

To Find

Power P.

Formula Used

$$P = \vec{F} \cdot \vec{v} = F_x v_x + F_y v_y.$$

Solution

$$P = (7)(2) + (-3)(3).$$
$$P = 14 - 9.$$
$$\boxed{P = 5\ \mathrm{W}}$$

Unit check: $\mathrm{N} \times \mathrm{m\,s^{-1}} = \mathrm{J\,s^{-1}} = \mathrm{W}$.

40. Amplitude for 25 times the intensity

Question. To increase wave intensity by a factor of 25, calculate the required increase in amplitude.

Given Data

$$\frac{I_2}{I_1} = 25.$$

To Find

Amplitude ratio A_2/A_1.

Formula Used

For otherwise unchanged wave conditions:

$$I \propto A^2.$$

Therefore:

$$\frac{I_2}{I_1} = \left(\frac{A_2}{A_1}\right)^2.$$

Solution

$$25 = \left(\frac{A_2}{A_1}\right)^2.$$

Take square roots:

$$\frac{A_2}{A_1} = \sqrt{25} = 5.$$

$$\boxed{A_2 = 5A_1}$$

The amplitude must **become five times its original value**.

The actual increase is:

$$A_2 - A_1 = 5A_1 - A_1 = 4A_1.$$

Percentage increase:

$$\frac{4A_1}{A_1}(100) = 400\%.$$

Understanding: A factor of 25 in intensity requires a factor of $\sqrt{25} = 5$ in amplitude, not a factor of 25.

41. Entropy change when ice melts

Question. Calculate the entropy change when 800 g of ice melts at 0°C. The latent heat of fusion is 3.36×10^5 J/kg. Also explain the sign of the entropy change.

Given Data

$$m = 800 \text{ g} = 0.800 \text{ kg}.$$

$$T = 0 + 273 = 273 \text{ K}, \qquad L_f = 3.36 \times 10^5 \text{ J kg}^{-1}.$$

To Find

Entropy change ΔS and its sign.

Formula Used

$$Q = mL_f, \qquad \Delta S = \frac{Q_{\text{rev}}}{T}.$$

Step 1: Heat absorbed

$$Q = 0.800(3.36 \times 10^5).$$

$$Q = 2.688 \times 10^5 \text{ J}.$$

Step 2: Entropy change

$$\Delta S = \frac{2.688 \times 10^5}{273}.$$

$$\boxed{\Delta S \approx +985 \text{ J K}^{-1}}$$

Sign: Positive. The ice absorbs heat and becomes liquid water, which has greater entropy at the melting point.

For unseen questions: Melting uses positive heat for the substance; freezing uses negative heat. Always use temperature in kelvin in the entropy formula.

42. Angular speed for artificial gravity

Question. A spaceship of diameter 8 m rotates about its axis to produce artificial gravity of magnitude 17.47 m/s^2. Find its angular speed.

Given Data

$$d = 8\ \text{m}, \qquad r = \frac{d}{2} = 4\ \text{m}.$$

$$a_c = 17.47\ \text{m s}^{-2}.$$

To Find

Angular speed ω.

Formula Used

$$a_c = \frac{v^2}{r}, \qquad v = r\omega.$$

Combining:

$$a_c = \frac{(r\omega)^2}{r} = r\omega^2.$$

Solution

$$\omega^2 = \frac{a_c}{r}.$$

$$\omega = \sqrt{\frac{17.47}{4}} = \sqrt{4.3675}.$$

$$\boxed{\omega = 2.09\ \text{rad s}^{-1}}$$

Common mistake: Use radius 4 m, not diameter 8 m. Here the required artificial gravity is supplied by the centripetal acceleration at the rim.

43. Pendulum period and uncertainty

Question. Ten vibrations of a simple pendulum take 17.6 s. The stopwatch least count is 0.1 s. Find the time period and its uncertainty.

Given Data

$$t = 17.6 \text{ s}, \qquad N = 10.$$

Taking the stated least count as the uncertainty in the total measured time:

$$\Delta t = 0.1 \text{ s}.$$

To Find

Time period T and uncertainty ΔT.

Formula Used

$$T = \frac{t}{N}.$$

Since the number of vibrations is counted exactly:

$$\Delta T = \frac{\Delta t}{N}.$$

Solution

$$T = \frac{17.6}{10} = 1.76 \text{ s}.$$

$$\Delta T = \frac{0.1}{10} = 0.01 \text{ s}.$$

$$\boxed{T = (1.76 \pm 0.01) \text{ s}}$$

Percentage uncertainty:

$$\frac{\Delta T}{T}(100) = \frac{0.01}{1.76}(100).$$

$$\boxed{\text{Percentage uncertainty} \approx 0.57\%}$$

Understanding: Timing several vibrations reduces the uncertainty in the calculated period compared with timing a single vibration using the same absolute timing uncertainty.

44. Circuit with 10 V and 6 V cells

Question. Find the current through each resistor of the following circuit.

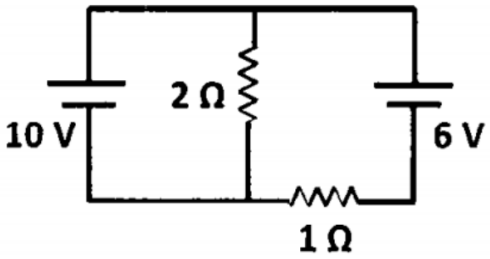


Figure 2: Circuit supplied in the question.

Given Data

$$\varepsilon_1 = 10\ \text{V}, \quad \varepsilon_2 = 6\ \text{V}, \quad R_1 = 2\ \Omega, \quad R_2 = 1\ \Omega.$$

Both cells have their positive terminals at the top.

To Find

Current in each resistor, with direction.

Laws Used

$$V = IR, \qquad \sum \Delta V = 0.$$

Step 1: Current through the 2 ohm resistor

The 10 V cell is directly across this resistor:

$$I_{2\Omega} = \frac{10}{2}.$$

$$\boxed{I_{2\Omega} = 5.0\ \text{A downward}}$$

Step 2: Current through the 1 ohm resistor

Take the bottom-left node as zero potential. The 10 V cell gives:

$$V_{\text{top}} = 10\ \text{V}.$$

For the 6 V cell:

$$V_{\text{top}} - V_{\text{bottom-right}} = 6.$$

$$V_{\text{bottom-right}} = 10 - 6 = 4\ \text{V}.$$

Thus the 1 ohm resistor has 4 V across it:

$$I_{1\Omega} = \frac{4 - 0}{1}.$$

$$\boxed{I_{1\Omega} = 4.0\ \text{A from right to left}}$$

Equivalently, the outer-loop equation is $10 - 6 - I_{1\Omega}(1) = 0$.

Check: The 10 V source supplies $5 + 4 = 9$ A at the top junction.

45. Proton mass doubled

Question. At what speed would the mass of a proton be doubled? Its rest mass is 1.673×10^{-27} kg.

Given Data

$$m = 2m_0, \qquad m_0 = 1.673 \times 10^{-27}\ \text{kg}.$$
$$c = 3 \times 10^{8}\ \text{m s}^{-1}.$$

To Find

Speed v.

Formula Used

Using the relativistic-mass convention in the question:

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}.$$

Solution

$$2m_0 = \frac{m_0}{\sqrt{1 - v^2/c^2}}.$$

Cancel m_0:

$$2 = \frac{1}{\sqrt{1 - v^2/c^2}}.$$
$$\sqrt{1 - v^2/c^2} = \frac{1}{2}.$$

Square both sides:

$$1 - \frac{v^2}{c^2} = \frac{1}{4}.$$
$$\frac{v^2}{c^2} = \frac{3}{4}.$$
$$v = c\sqrt{\frac{3}{4}} = (3 \times 10^{8})(0.866025).$$
$$\boxed{v = 2.60 \times 10^{8}\ \text{m s}^{-1} \approx 0.866c}$$

Check: The result is below c, as required for a proton.

46. Current through the central 1 ohm resistor

Question. Calculate the current through the 1 ohm resistor in the following circuit.

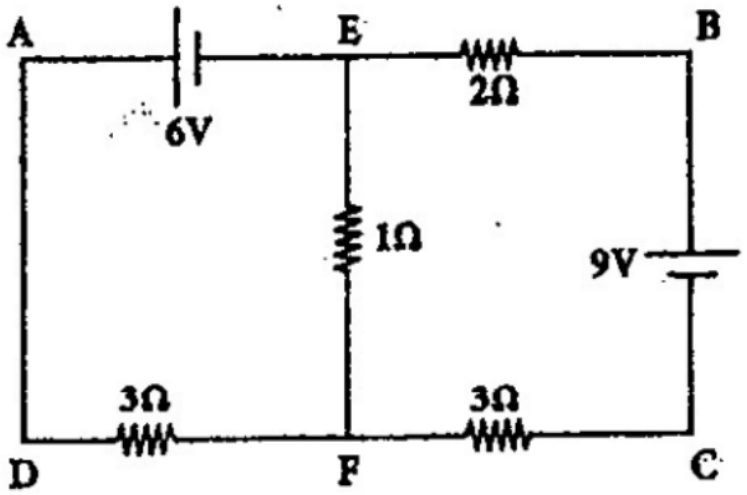


Figure 3: Circuit supplied in the question.

Given Data

Left cell: 6 V, positive terminal toward A. Right cell: 9 V, positive terminal toward B.

$$R_{DF} = 3\,\Omega, \quad R_{EB} = 2\,\Omega, \quad R_{FC} = 3\,\Omega, \quad R_{EF} = 1\,\Omega.$$

To Find

Current through EF and its direction.

Laws Used

$$\sum I_{\text{in}} = \sum I_{\text{out}}, \qquad \sum \varepsilon - \sum IR = 0.$$

Step 1: Assume current directions

Let I_L flow through the left branch $E \to A \to D \to F$.

Let I_R flow through the right branch $E \to B \to C \to F$.

Let I flow upward through the central resistor, $F \to E$.

At junction F:

$$\boxed{I = I_L + I_R} \tag{1}$$

An assumed current may turn out negative. That means its actual direction is opposite.

Step 2: Left-loop equation

Travel around $E \to A \to D \to F \to E$:

$$6 - 3I_L - I(1) = 0.$$

$$3I_L + I = 6.$$

Substitute $I = I_L + I_R$:

$$\boxed{4I_L + I_R = 6} \tag{2}$$

Question 46 - continued

Step 3: Right-loop equation

Travel around $E \to B \to C \to F \to E$. The 9 V cell is crossed from positive to negative:

$$-2I_R - 9 - 3I_R - I = 0.$$

$$5I_R + I = -9.$$

Substitute Equation (1):

$$\boxed{I_L + 6I_R = -9} \tag{3}$$

Step 4: Solve simultaneously

From Equation (2):

$$I_R = 6 - 4I_L.$$

Substitute into Equation (3):

$$I_L + 6(6 - 4I_L) = -9.$$

$$I_L + 36 - 24I_L = -9.$$

$$-23I_L = -45.$$

$$I_L = \frac{45}{23}\ \text{A}.$$

Then:

$$I_R = 6 - 4\left(\frac{45}{23}\right) = -\frac{42}{23}\ \text{A}.$$

The negative sign means the right-branch current actually flows opposite to the assumed direction.

Finally:

$$I = I_L + I_R = \frac{45}{23} - \frac{42}{23} = \frac{3}{23}\ \text{A}.$$

$$\boxed{I = 0.130\ \text{A upward, from F to E}}$$

For unseen questions: Choose current directions consistently, write the signs carefully, and let the algebra determine the actual direction.

47. Time taken by an accelerating missile

Question. A missile's speed increases from 1500 m/s to 5000 m/s after travelling 70 km. Calculate the time taken to reach this speed.

Given Data

$$u = 1500\ \mathrm{m\,s^{-1}}, \qquad v = 5000\ \mathrm{m\,s^{-1}}.$$

$$s = 70\ \mathrm{km} = 70000\ \mathrm{m}.$$

Assume constant acceleration.

To Find

Time t.

Formula Used

$$v^2 - u^2 = 2as, \qquad v = u + at.$$

Step 1: Find acceleration

$$a = \frac{v^2 - u^2}{2s}.$$

$$a = \frac{5000^2 - 1500^2}{2(70000)}.$$

$$a = \frac{25000000 - 2250000}{140000}.$$

$$a = \frac{22750000}{140000} = 162.5\ \mathrm{m\,s^{-2}}.$$

Step 2: Find time

$$t = \frac{v - u}{a} = \frac{5000 - 1500}{162.5}.$$

$$t = \frac{3500}{162.5}.$$

$$\boxed{t \approx 21.5\ \mathrm{s}}$$

Quick Check

For constant acceleration:

$$v_{\text{average}} = \frac{u + v}{2} = \frac{1500 + 5000}{2} = 3250\ \mathrm{m\,s^{-1}}.$$

$$t = \frac{s}{v_{\text{average}}} = \frac{70000}{3250} = 21.54\ \mathrm{s}.$$

48. Entropy change when heat is removed

Question. A body at 200°C undergoes a reversible isothermal process. The heat removed is 7300 J. Calculate its entropy change and explain the sign.

Given Data

$$T = 200 + 273 = 473\ \text{K}.$$

Heat is removed from the body, so:

$$Q = -7300\ \text{J}.$$

The process is reversible and isothermal.

To Find

Entropy change ΔS and its sign.

Formula Used

$$\Delta S = \frac{Q_{\text{rev}}}{T}.$$

Solution

$$\Delta S = \frac{-7300}{473}.$$

$$\boxed{\Delta S = -15.4\ \text{J K}^{-1}}$$

Sign: Negative, because the body loses heat and its entropy decreases.

For reversible heat transfer, the surroundings gain an equal amount of entropy:

$$\Delta S_{\text{surroundings}} = +15.4\ \text{J K}^{-1}.$$

Therefore:

$$\Delta S_{\text{total}} = 0.$$

For unseen questions: Heat entering the body is positive; heat leaving it is negative. The entropy change of the body and that of the surroundings are different quantities.

How to approach an unseen numerical

First identify what is happening physically. Then choose the formula that matches that condition.

Condition or wording	Useful starting point
Highest point of vertical motion	$v = 0$, but $a = -g$ if upward is positive
No energy loss	$KE_i + PE_i = KE_f + PE_f$
Friction or air resistance	Include its signed work in the energy balance
Connected pipes	$A_1 v_1 = A_2 v_2$
Sound source moving away	Observed frequency decreases
Floating in equilibrium	Net force is zero
Circular motion in a magnetic field	Magnetic force supplies centripetal force
Power with vector components	$P = \vec{F} \cdot \vec{v}$
Melting or freezing	$Q = mL$, then $\Delta S = Q_{\text{rev}}/T$
Circuit with several branches	Label currents and apply junction and loop laws

Useful Unit Conversions

$$1\ \text{g} = 10^{-3}\ \text{kg}, \qquad 1\ \text{mm} = 10^{-3}\ \text{m}.$$

$$1\ \text{mm}^2 = 10^{-6}\ \text{m}^2, \qquad 1\ \mu\text{C} = 10^{-6}\ \text{C}.$$

$$1\ \text{km} = 10^{3}\ \text{m}, \qquad T(\text{K}) = t(^\circ\text{C}) + 273.$$

Before Submitting an Exam Answer

Check that you have used the radius rather than the diameter where required, squared areas and distances correctly, converted temperature to kelvin, shown signs consistently, and included the final unit. For a vector quantity, add its direction.

Notes About the Supplied Questions

Questions 3 and 5 omit data needed for unique numerical answers. Question 7 prints an incorrect sphere surface-area formula. Question 12 has an apparent symbol error. Question 24 permits two angles when its absolute-value notation is taken literally. Question 25 has been solved using the printed resistivity exponent. Question 32 gives the Carnot maximum when only the reservoir temperatures are known. Questions 1 and 38 state the modelling assumptions needed for their intended calculations.

Made by Fakhar STEM Sphere